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SUPERSEDING
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MILITARY SPECIFICATION

ANESTHESIA APPARATUS, SPINAL, TUOHY

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 This specification covers a spinal anesthesia apparatus of Tuohy design, for use in administration of continuous spinal anesthesia.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

GG-I-526—Instruments, Dental and Surgical; General Specifications.

GG-N-196—Needles, Hypodermic; for Luer syringes.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking of Shipments.

DRAWINGS

Armed Services Medical Procurement Agency 1743—Spinal Anesthesia Apparatus, Tuohy.

(Reduces copies of drawing on attached as figures 1 and 2.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 **Material.**

3.1.1 *Metal parts.*—All metal parts of the Tuohy spinal anesthesia apparatus shall conform to Specification GG-I-526, the following classes and compositions of materials applying for the various metal component parts of the apparatus:

1. Needle and needle obturator canula—class 2, composition G (c), except that Rockwell hardness shall be C-30 to C-40.
2. Needle hub and obturator handle—class 4, composition N, except that nickel plating only or chromium over nickel plating shall be used.
3. Stopcock, catheter adapter, and adapters for rubber tubing—class 4, composition N, except that nickel plating only or chromium over nickel plating shall be used.
4. Catheter stylet—class 2, composition G (c).

3.1.2 *Catheter.*

3.1.2.1 *Fabric.*—Fabric shall be woven from high tenacity nylon yarn, or other suitable material as approved by the contracting officer.

3.1.2.2 *Coating.*—Coating shall be a flexible varnish, or equal, containing lead or other X-ray opaque material.

3.1.3 *Washer for catheter adapter.*—Washer for catheter adapter shall be natural rubber compound of approximately 65 percent by volume rubber hydrocarbon.

3.2 **Construction.**—The Tuohy spinal anesthesia apparatus shall consist essentially of the following components:

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(a) Spinal anesthesia needle.
(b) Graduated X-ray opaque woven catheter.
(c) Catheter adapter.
(d) Washer for catheter adapter.
(e) Three-way stopcock.
(f) Two adapters for rubber tubing.

3.2.1 *Needle.*—It shall be a spinal needle, Tuohy type, with Huber point. Needle shall have a 16-gage canula, with outside diameter of 0.065 inch and inside diameter of 0.047 inch, and shall be provided with an obturator. Needle and obturator shall conform with illustration for Stock No. 3–025–250 as shown on drawing.

3.2.2 *Catheter.*

3.2.2.1 *Description.*—Catheter shall be of the graduated X-ray woven type of administration of continuous spinal anesthesia. It shall be composed of a tightly and smoothly woven base, free from loose ends or broken fibers, and completely, uniformly, and symmetrically covered with a flexible coating containing the X-ray opaque material. The finished catheter shall have proper flexibility, pliability, and body required to withstand the pressure of introduction through the 16-gage needle with the softness and pliability necessary to prevent irritation or injury in the spinal canal. Catheter shall be 35 to 36 inches in over-all length, size 3.5 French, woven on a mandrel 0.020 inch in diameter. Catheter shall be capable of passing freely through the 16-gage spinal needle specified in 3.2.1. The lumen of the catheter shall be such, that after being tested as specified in 4.3.1.1, an 0.018-inch mandrel shall be capable of being passed completely through the catheter. The tip, or distal end, of the catheter shall have a plain open end with smooth beveled edges, and shall have no eyes or side openings.

3.2.2.2 *Base.*—The base shall be a single web, tightly and smoothly woven in braided form with yarn of appropriate size and type for the size of the tube on a 26 to 34 carrier machine. The thickness of the wall shall be relatively uniform throughout the length of the tube. Each yarn woven into the tube shall be of one continuous length from one end of the tube to the other. The tube shall have the required body to secure resistance to buckling or breaking when tested as specified in 4.3.1.1, 4.3.1.3, 4.3.1.4, and 4.3.1.5.

3.2.2.3 *Coating.*—The coating shall be thoroughly and uniformly impregnated in and around the woven base in a manner to coat the braided fibers completely. The coating thickness shall be relatively uniform throughout the length of the catheter. The coating shall be formulated to secure maximum adhesion to the tube and shall have the required toughness and elasticity to permit repeated flexing and to prevent peeling, cracking, or breaking when tested as specified in 4.3.1.3, 4.3.1.4, and 4.3.1.5. The coating shall be resistant to body fluids, mild acids, alkalies, and deterioration due to aging during storage. The coated catheter shall show no signs of brittleness, blistering, or other visible damage as a result of the boiling or autoclaving tests specified in 4.3.1.3 and 4.3.1.4. The coated catheter shall not be tacky, shall not stick together, and its surface shall not be marred or show any other evidence of damage as a result of the flexibility tests specified in 4.3.1.5.

3.2.2.4 *Toxicity.*—The catheters shall be free from toxic material or any ingredient that might have a deleterious effect upon body tissues.

3.2.2.5 *X-ray opacity.*—The catheters shall throw a definite, uniform shadow, when tested as specified in 4.3.1.6. The degree of opaqueness shall be such that the shadow of a small foreign body in the front or rear of the catheter shall be visible for diagnosis.

3.2.2.6 *Finish.*—The surface of each catheter shall be properly worked down and given a dry, smooth, highly polished but elastic finish, free from scratches, grooves, or other defects which would prevent its easy passage through the 16-gage spinal needle or the withdrawal of the needle over the catheter after its insertion into the spinal canal. The surface shall be "slick" when wet and pass easily.

3.2.2.7 *Stylet.*—Each catheter shall be furnished with a stylet of proper size to pass freely through the full length of the lumen.

3.2.2.8 *Identification marking.*—Each cathe-

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ter shall be plainly and indelibly marked upon the proximal end with manufacturer's name or registered trade-mark, "3.5" (size according to French scale) and the word "X-Ray." Each catheter shall be graduated in the metric system, based upon groupings of 1, 5, and 25 centimeters. These graduations shall be legibly and indelibly marked.

3.2.3 *Catheter adapter.*—It shall be a Tuohy luer-lock catheter adapter consisting of a male luer-lock connecting piece, a catheter holding piece, and a rubber washer. Catheter adapter shall conform with illustration for Stock No. 3–025–200 as shown on drawing. Each adapter shall be supplied with two additional washers as specified in 3.2.4.

3.2.4 *Washer for catheter adapter.*—The washer for the catheter adapter shall measure three-sixteenths inch in length (plus or minus tolerance of one sixty-fourth inch), three thirty-seconds inch inside diameter (plus or minus tolerance of one sixty-fourth inch), and one thirty-second inch in wall thickness (plus tolerance of one sixty-fourth inch). The washer shall conform with illustration for Stock No. 3–025–275 as shown on drawing.

3.2.5 *Three-way stopcock.*—It shall be a three-way stopcock for use in controlling the flow of solutions during anesthesia procedures, and shall be in accordance with illustration for Stock No. 3–744–760 as shown on drawing. The three connectors of the stopcock shall consist of female luer-lock connection, a male luer-lock connection, and connection capable of accommodating rubber tubing of one-eighth inch inside diameter and one thirty-second inch thick wall.

3.2.6 *Adapters for rubber tubing.*—Adapters shall be of two styles, one style for connecting rubber tubing to a syringe and the other style for connecting rubber tubing to the stopcock. Both styles shall be so constructed that one end of the adapter shall be capable of accommodating rubber tubing having a one-eighth inch inside diameter and a one thirty-second inch thick wall. The styles shall differ in design as specified below.

3.2.6.1 *Adapter, tubing to syringe.*—Adapter shall conform with illustration for Stock No. 3–000–720 as shown on drawing.

3.2.6.2 *Adapter, tubing to stopcock.*—Adapter shall conform with illustration for Stock No. 3–000–500 as shown on drawing.

3.3 **Fluid tightness.**—The three adapters described above, the spinal anesthesia needle, and the 3-way stopcock shall be capable of passing test for fluid tightness specified in 4.3.2.

3.4 **Workmanship.**—Workmanship shall be first class throughout. Spinal anesthesia apparatus shall be free from defects which detract from its appearance or impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.1). In addition, inspection at contractor's facility by inspector shall be conducted in accordance with Standard MIL-STD-105.

4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Tests.**—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

4.3.1 *Catheter.*

4.3.1.1 *Mandrel test.*—Each sample catheter shall be immersed for 3 hours in distilled water at room temperature. An 0.018-inch mandrel shall then be passed completely through the catheter.

4.3.1.2 *Outside diameter.*—The outside diameter of the catheters selected for test shall be checked for French scale catheter dimensions and also by passing the catheter through the 16-gage spinal needle specified in 3.2.1.

4.3.1.3 *Boiling test.*—Each sample catheter

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shall be boiled for ten 20-minute intervals, with 24 hours between each boiling to permit drying out at room temperature (68° to 72° F.).

4.3.1.4 *Autoclaving test.*—Each sample catheter selected for test shall be dry-autoclaved for ten 10-minute intervals at 250° F., with sufficient time between to permit cooling to room temperature (68° to 72° F.).

4.3.1.5 *Flexibility tests.*

4.3.1.5.1 *Wet test.*—The catheter shall be twisted three full-length turns around a catheter of the same size, subjected to boiling for 20 minutes, allowed to return to room temperature, then untwisted.

4.3.1.5.2 *Dry test.*—The catheter shall be twisted three full-length turns around a catheter of the same size, subjected to autoclaving for 15 minutes at 250° F., allowed to return to room temperature, then untwisted.

4.3.1.6 *X-ray opacity.*—Each catheter shall be radiographed as follows:

> 85 KVP, 10 Milliampere-seconds, 25-inch distance, 24 centimeters of U. S. P. paraffin superimposed over catheters, and using a standard make of cassette with screens, films, and processing solutions.

The foreign body used for test purposes shall be a small fragment of sheet lead one-half millimeter in thickness.

4.3.2 *Fluid tightness.*

4.3.2.1 *Stopcock and epinal needle.*—The spinal needle shall be fitted to the stopcock and, with the needle and stopcock in horizontal position, a hub of the stopcock shall be attached by means of a standard luer taper gage to a water pressure source, having a pressure gage. The water shall be run through the stopcock so as to fill any cavity space and the lumen of the luer lock and the needle. The lumen of the needle shall be sealed by a piece of solid rubber slipped on to the point, and a pressure of 45 pounds square inch applied for 10 seconds. Neither the swaged connection of the canula with the hub of the needle, nor the luer connection of the hub of the needle with the male luer lock of the stopcock, and no part of the stopcock nor any of the connection points of the stopcock shall show leakage exceeding the formation of a drop.

4.3.2.2 *Catheter adapter.*—An X-ray woven catheter shall be fitted to the catheter attachment end of the catheter adapter. The female luer hub of the catheter adapter shall be fitted (in horizontal position) by means of a standard luer taper gage to a water pressure source, having a pressure gage. The water shall be run through the adapter so as to fill any cavity space of the adapter and the lumen of the catheter. The free end of the catheter shall be sealed and a pressure of 45 pounds per square inch applied for 10 seconds. The luer connection of the hub of the adapter to the taper gage, and the connection of the catheter to the adapter, shall show no leakage exceeding the formation of a drop.

4.3.2.3 *Adapter, tubing to syringe.*—Fluid tightness test for the tubing to syringe adapter shall be identical to that for the catheter adapter described in 4.3.2.2, except that here the tubing attachment end shall be sealed.

4.3.2.4 *Adapter, tubing to stopcock.*—A hypodermic needle conforming to style 1 or 2 hub of Specification GG-N-196 shall be attached to the luer slip tip of the tubing to stopcock adapter and the adapter and needle, in horizontal position, attached to a water pressure source, having a pressure gage. The water shall be run through the needle so as to fill the lumen of the needle and the luer slip and any cavity space of the adapter. The tubing attachment end of the adapter shall be sealed and a pressure of 45 pounds per square inch applied for 10 seconds. There shall be no leakage between the needle hub and the luer slip tip of the adapter exceeding the formation of one drop.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.—Packaging, packing, and marking shall be as specified in applicable purchase description.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sampling required (see 4.1).
(c) Inspection required (see 4.2).
(d) Place of final inspection (see 4.2).

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6.2 This specification covers the following items appearing in the Armed Services Catalog of Medical Matériel:

3–000–500—Adapter, ⅛-Inch Tubing to Luer Slip Needle.
3–000–720—Adapter, Luer Slip Syringe to ⅛-Inch Tubing.
3–025–200—Adapter, Catheter, Spinal Anesthesia, Tuohy.
3–025–225—Catheter, Spinal Anesthesia, X-Ray, Woven, Tuohy, 3½ Fr.
3–025–250—Needle, Spinal Anesthesia, Tuohy, 16-Gage, 3-Inch.
3–025–275—Washer, Catheter, Adapter, Rubber, 12s.
3–744–760—Stopcock, 3-Way, Luer Slip Syringe, Luer Lock Needle and Tubing Connections.

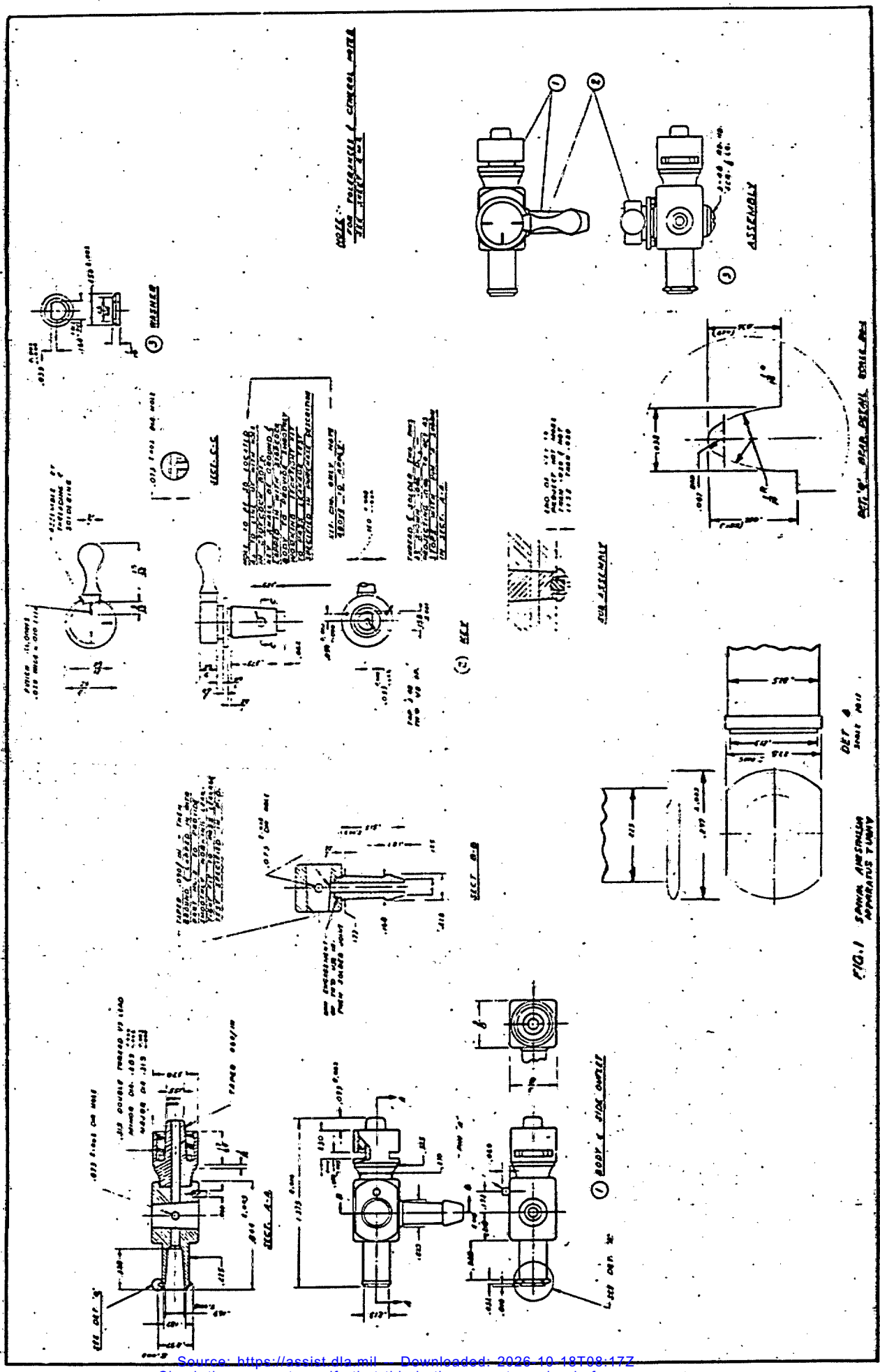
6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet military requirements.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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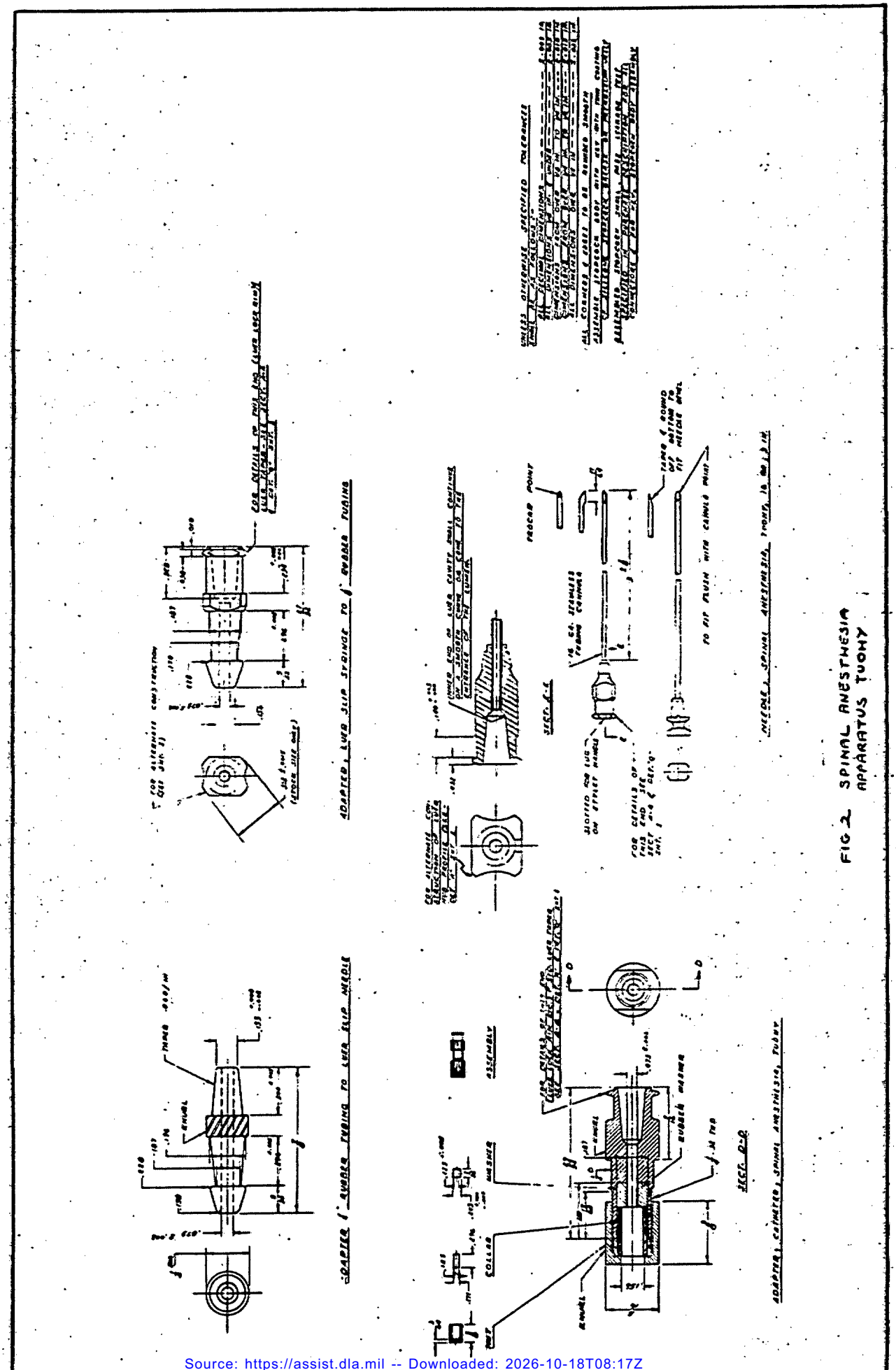


FIG 2 SPINAL ANESTHESIA
APPARATUS TUOHY

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